



Does conditioning leave a structural memory?

Large-area structural evidence for the dislocation picture of conditioning

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2026-09-28

Does conditioning leave a structural memory?

This is the sequel to last year's MeVArc talk. In 2025 I showed our dislocation model of conditioning and cross-sectional STEM that found a modified near-surface layer, and I closed by asking for more comparative forensics—other materials, more samples. This year answers part of that: a large-area copper measurement that tests the model's central prediction, and a stainless-steel experiment that will test whether the effect generalises. E_S is deliberately kept out of the introduction; it appears once, near the end.

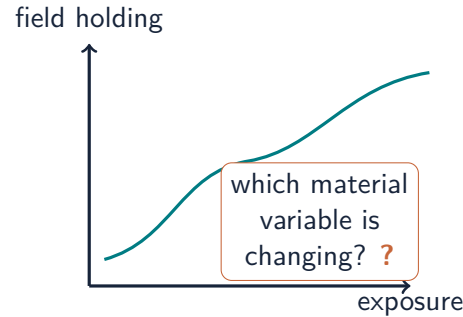


Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.
- ▶ The field itself drives conditioning, not the arcs.

Something in the metal stores the improvement. We do not know what.



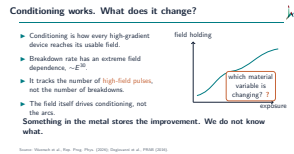
Source: Wuensch et al., Rep. Prog. Phys. (2026); Degiovanni et al., PRAB (2016).

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└ Conditioning works. What does it change?

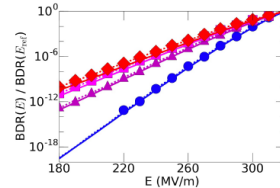
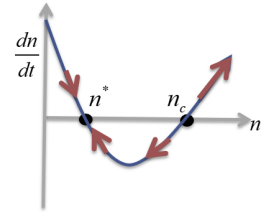
Open on the operational fact and the puzzle. Do not introduce E_S here. Two macroscopic clues already constrain the answer: the near-thirtieth-power field dependence, and that conditioning follows pulse count rather than breakdown count. Both say the applied field, not the arc, is doing the work—so we should look for a field-driven change in the metal itself.



Our model: conditioning as collective dislocation dynamics



- Field holding rises FCC→BCC→HCP — the order of barriers to dislocation motion.
- The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.
- One model gets the $\sim E^{30}$ rate, its temperature dependence, and the dark-current spikes.



Central prediction: conditioning *evolves the near-surface dislocation structure.*

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└ Our model: conditioning as collective dislocation dynamics

This is our own model (2018–2020). The cartoon shows the birth–death dynamics of the mobile-dislocation population: below a critical density the population decays, above it a runaway triggers nucleation. The plot is the reproduced field dependence of the breakdown rate. The key line for this talk is the prediction highlighted at the bottom: conditioning should leave an evolving dislocation structure just below the surface. Everything after this slide is a test of that sentence.

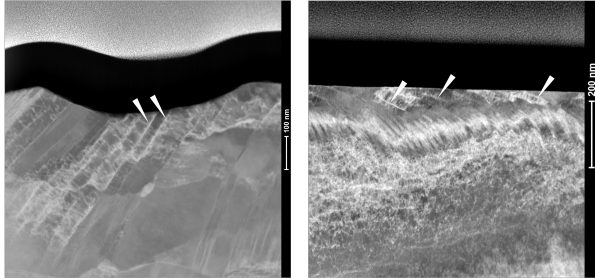
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Central prediction: conditioning evolves the near-surface dislocation structure.

Source: Engelberg et al., PRL 120, 124001 (2018); PRAB 22, 083501 (2019); PRAB 23, 123501 (2020).

STEM already saw it, at single spots



Cross-sectional STEM of conditioned hard (as-machined) Cu: dislocation-denuded zone in the top ~ 200 nm.

- ▶ First **direct** structural evidence of field-driven near-surface plasticity.
- ▶ Seen at both 300 K and 30 K.
- ▶ Fits the macroscopic clue: soft Cu conditions far more slowly than hard Cu, so the starting dislocation population matters.
- ▶ Hard lesson: FIB preparation plants its own dislocations. Trust only large-scale structure.
- ▶ One cross-section cannot speak for the whole electrode.

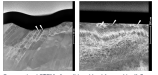
Confirmed at one spot. The large-area comparison was still missing.

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Does conditioning leave a structural memory?

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Confirmed at one spot. The large-area comparison was still missing.

This is our 2025 STEM work. The dark band is the protective coating over the surface; just beneath it the field-exposed region is denuded of dislocations, while the reference stays dislocation-rich to the surface. The hard-versus-soft copper contrast (Korsbäck 2020) is an independent macroscopic clue pointing the same way: about 25 million pulses to condition hard copper versus roughly 600 million for soft, consistent with different starting dislocation populations. Two lessons carried forward: FIB sample preparation itself generates dislocations and loops—real red herrings—so only larger-scale structure is trustworthy; and TEM/STEM is intrinsically local. Last year I closed by asking for more comparative forensics: other materials, more samples. That is exactly the gap the next slides fill.

A tiny stress, repeated a billion times



Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: **0.028 MPa**

$\sim 10^3 \times$ below the yield of annealed Cu

- ▶ Fatigue at 10^8 – 10^{10} cycles rearranges dislocations irreversibly, well below yield.
- ▶ Here: $\sim 10^9$ conditioning pulses.
- ▶ Pulsed DC means **no RF pulsed heating** and no bulk currents. Maxwell stress is the only cyclic load.
- ▶ MDDF is **threshold-free**: no minimum stress amplitude.

The model predicts a large-area signature. Last year we asked for it; this year we looked.

Source: Mughrabi, MMTB 40, 431 (2009); Stanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007).

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Does conditioning leave a structural memory?

└ A tiny stress, repeated a billion times

Head off the obvious objection: the Maxwell stress is three orders of magnitude below yield. The point is not amplitude but accumulation. VHCF shows that oscillatory loading well below yield still rearranges dislocations irreversibly when sustained for a billion cycles, which is our regime. The pulsed-DC system is also a cleaner experiment than RF structures: there is no RF magnetic field, no pulsed surface heating, and no bulk Ohmic current, so the Maxwell stress is the only cyclic mechanical load—earlier RF-based studies could never fully separate thermal fatigue from field effects. And the MDDF description is threshold-free, so it does not need the stress to exceed any yield-like value. This is the bridge from the 2025 model-and-microscopy story to this year's large-area measurement.

A tiny stress, repeated a billion times

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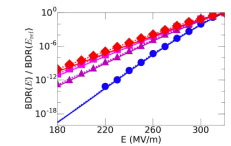
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Last year we asked for more forensics. Here it is.

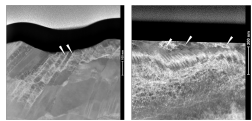


MeVArc 2025



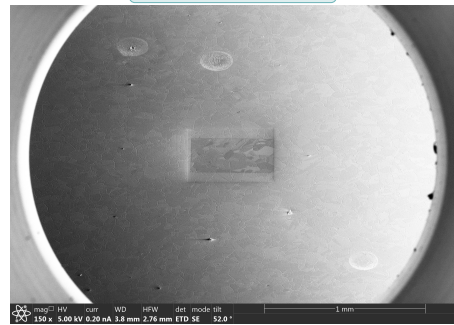
MDDF model

Closed with: other materials? more samples?



local STEM denuded zone

MeVArc 2026

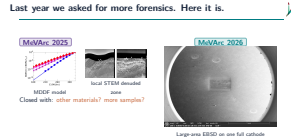


Large-area EBSD on one full cathode

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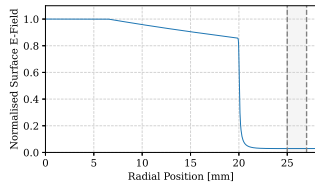
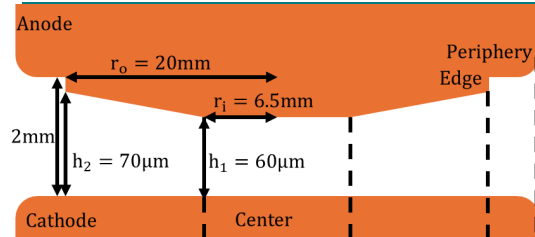
Does conditioning leave a structural memory?

└ Last year we asked for more forensics. Here it is.



Light visual bridge: 2025 showed the model and a local cross-section; 2026 delivers the large-area structural comparison on one electrode. Do not quote the 75 percent effect here.

One cathode turns field exposure into a controlled coordinate



- OFE Cu, pulsed DC
- 1 μ s pulses at 1 kHz
- Peak field ~ 80 MV/m at center
- Nine EBSD regions: center, edge, periphery, and external reference

The field gradient gives us internal controls on a single specimen.

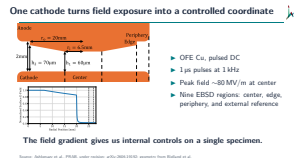
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; geometry from Bjelland et al.

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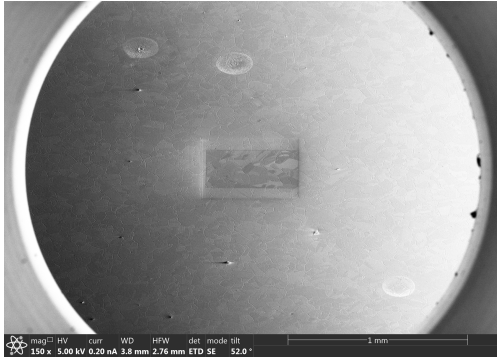
Does conditioning leave a structural memory?

One cathode turns field exposure into a controlled coordinate

The sloped anode maps radius onto local field: highest at the flat center, decaying across the sloped gap, and below about 2.5 MV/m beyond the anode radius. Comparing positions on one cathode removes cathode-to-cathode preparation differences, and a separate identically treated but unexposed cathode gives an external reference. This is the “more comparative forensics” design from last year’s outlook.



EBSD maps plastic activity far from breakdown craters



Cleaned ROI is hundreds of microns from visible craters.

- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step sets the lateral resolution only
- ▶ Information depth: a few tens of nm
- ▶ ~24,000 indexed points per ROI
- ▶ LAM, KAM, and LOS probe intragrain orientation gradients
- ▶ Misorientation is a proxy for GND density

We measure the cumulative response to the field, not crater damage.

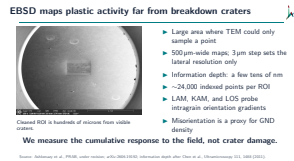
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; information depth after Chen et al., Ultramicroscopy 111, 1488 (2011).

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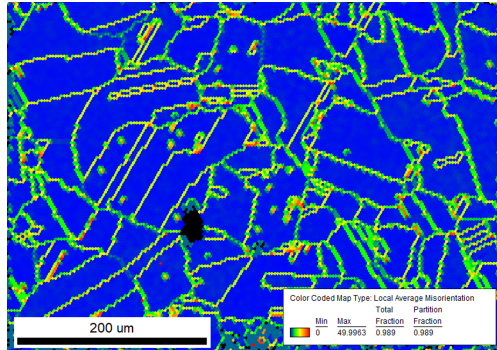
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└ EBSD maps plastic activity far from breakdown craters

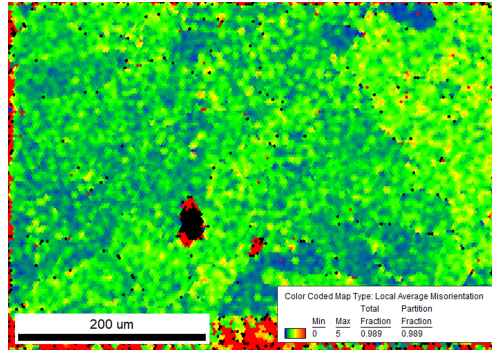
EBSD is chosen precisely because it is large-area where TEM is local. A shallow FIB clean removes oxide, under 10 nm by AFM, then we map orientation gradients. The 3 micron step and kernel fix only the lateral resolution; the backscattered electrons that carry the pattern come from the top few tens of nanometres of copper at 15 kV, so this is a near-surface measurement. All ROIs are deliberately far from craters, so we measure the cumulative material response to the field, not arc damage. LAM is the primary metric; KAM and LOS check that the result does not depend on one analysis choice.



How to read a LAM map: wide-angle vs low-angle



0–50° scale: grain boundaries dominate



0–5° scale: intragrain curvature

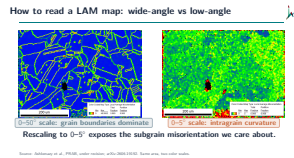
Rescaling to 0–5° exposes the subgrain misorientation we care about.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Same area, two color scales.

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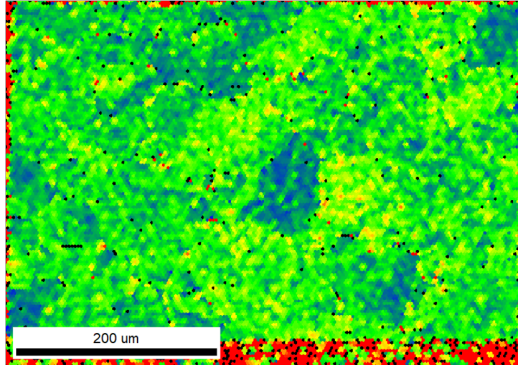
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How to read a LAM map: wide-angle vs low-angle

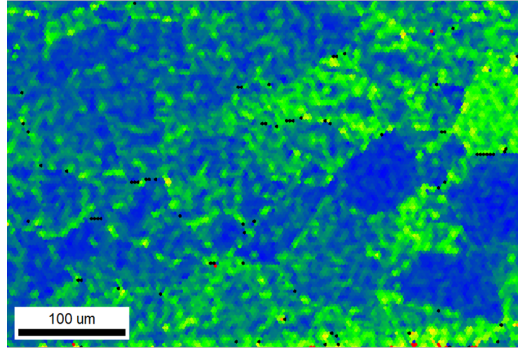


Teach the metric before the comparison. At the wide 0–50 degree scale only grain boundaries are visible. Rescaling the same map to 0–5 degrees reveals the intragrain lattice curvature—the low-angle misorientation that proxies geometrically necessary dislocations. Every LAM map that follows uses this same 0–5 degree scale.

High-field copper contains more intragrain curvature



Field-exposed center



Unexposed reference

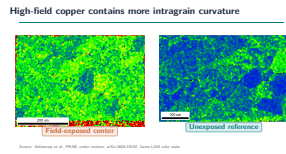
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Same LAM color scale.

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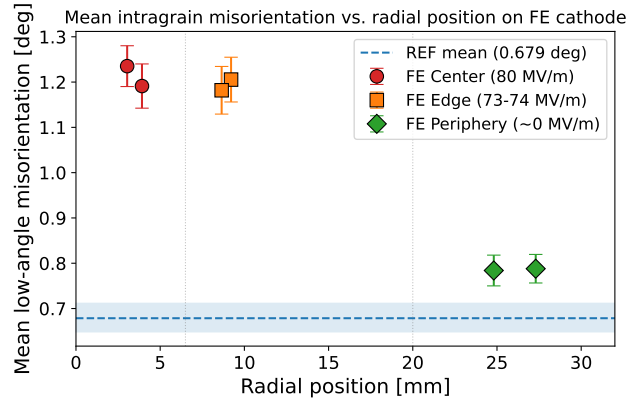
Does conditioning leave a structural memory?

└ High-field copper contains more intragrain curvature

Now the comparison. Look at the spatial texture, not just the average color: the field-exposed region carries far more high-misorientation structure inside the grains. This is one representative pair; the quantitative conclusion uses all nine ROIs.



The signal separates into three exposure tiers



- Center and edge: $\sim 1.2^\circ$
- External reference: $\sim 0.68^\circ$
- High field is $\sim 75\%$ higher
- Grain size does not follow this hierarchy

The ordering follows field history, not sample position or grain size.

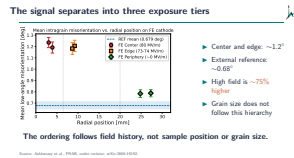
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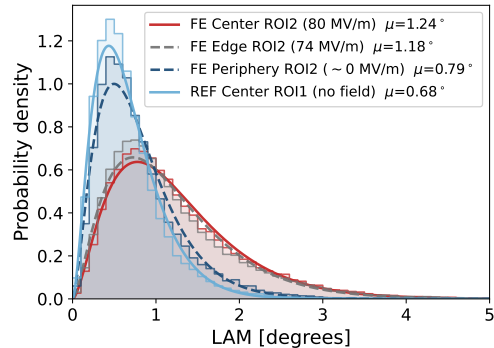
Does conditioning leave a structural memory?

The signal separates into three exposure tiers

Three tiers: high-field center and edge are highest and similar; the field-exposed periphery is intermediate; unexposed references are lowest. The periphery is powerful because it shared machining, heat treatment, mounting, vacuum, and handling with the high-field regions but saw at most about 2.5 MV/m—a low-field control, not a zero-field one. Its intermediate value is consistent with residual fringe-field exposure. The 75 percent figure characterises this cathode and history—not a universal constant.



Field exposure builds a high-misorientation tail



LAM tail above 2°

~0.14 field-exposed

~0.016 reference

~8× **increase**

Common gamma shape $k \approx 2.7$;

scale roughly doubles. All
center/edge vs unexposed:

$D_{KS} > 0.25$.

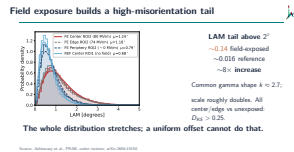
The whole distribution stretches; a uniform offset cannot do that.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

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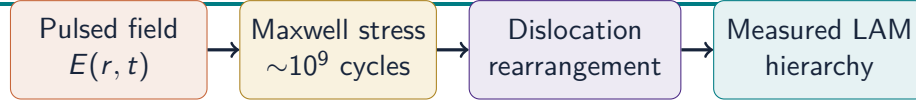
Does conditioning leave a structural memory?

Field exposure builds a high-misorientation tail



The mean is not the whole story. The low-angle distribution broadens and grows a long tail: the fraction above two degrees rises roughly eightfold. The gamma shape parameter stays near 2.7 while the scale roughly doubles—a fixed-shape stretch, the signature of progressive deformation. LAM, KAM, and LOS agree, and KS, chi-squared, and Anderson–Darling all separate the distributions. I quote effect sizes and grain-corrected errors rather than the extreme pixel-pair p-values.

The large-area test confirms the model's prediction



- First **large-area** confirmation that conditioning evolves the near-surface dislocation structure, as MDDF predicted.
- No conflict with the 2025 denuded zone: one mechanism, two starting populations. Dense as-machined Cu depletes at the surface; the heat-treated Cu here builds curvature.
- The three-tier ordering also tracks the conditioning variable E_S : a *candidate* structural basis for it.

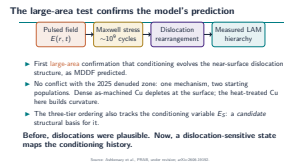
Before, dislocations were plausible. Now, a dislocation-sensitive state maps the conditioning history.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

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Does conditioning leave a structural memory?

└ The large-area test confirms the model's prediction



This is the payoff. The measured hierarchy is the large-area structural signature the model predicted and the STEM saw locally. The surface denuded zone and the EBSD curvature are not in conflict: they are one mechanism acting on two very different starting populations. In dense, as-machined copper the cyclic stress lets near-surface dislocations escape to the surface or annihilate, which is the depleted layer the TEM saw; in the heat-treated copper measured here there is little to deplete, so the same load generates and rearranges dislocations, which is the curvature EBSD records. The next slide says how the two pictures connect. Only now, and only in one line, do I connect to E_S : the spatial ordering matches the simulated E_S profile, so the dislocation population is a candidate structural basis for it. I never say we measured E_S .

interior field
reduced to E_0/ϵ_r

A diagram of a rectangular container divided by a horizontal dashed line. The top region is light blue and contains four green arrows pointing down and three orange arrows pointing up. The bottom region is light pink and contains several purple arrows pointing up.

$\sigma_M = \varepsilon_0 E^2/2$ enters
as elastic stress
denuded ~ 200 nm
 $\approx 8 \ell_D$ (hard Cu)
ensemble screens
stress
over $\ell_D =$
 $1/(4.2\sqrt{\rho}) \approx 25$ nm

Two-dimensional single-slip theory, $\rho \sim 10^{14} \text{ m}^{-2}$; no theory yet treats cyclic loading with a free surface, so the $\sim 200 \text{ nm}$ thickness is observed, not predicted. Not to scale.

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- └ Elastic screening: dislocations respond like bound charge

Picture the dislocation population as the bound charge of a dielectric: the Maxwell traction is the applied field, the dislocations rearrange in response, and in Groma's Debye-like theory the internal stress of any depleted or enriched layer is screened by the surrounding ensemble over about 25 nanometres for as-machined copper. Note the contrast on the right: the electric field itself is screened by surface charge within a nanometre and never enters the metal, whereas the load enters as elastic stress, so the electrostatic screening length says nothing about depth. The STEM denuded zone in hard copper is about eight of these screening lengths thick; no theory yet treats a cyclically loaded free surface, so that thickness is an observation, not a prediction.

Elastic screening: dislocations respond like bound charge

Dielectric in an applied field

Copper under the Maxwell traction

applied field $\rightarrow$ Maxwell traction + bound charge $\rightarrow$ dislocations (dipole regions) - Delay length $\sim \ell_D$ - increased interior field $\rightarrow$ interior stress of the displaced layer

Two-dimensional single-slip theory, $\mu \sim 10^{10} \text{ m}^{-2}$, as theory yet treats cyclic loading with a free surface, as in -300 nm thickness is observed, not scale to scale.

The field is screened by charge at the surface. The stress is screened by dislocations below it.

Copper establishes the effect, not universality



What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Confirms the MDDF prediction

What remains open

- ▶ One conditioned cathode / one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

Call for cooperation: more cathodes, other materials, other facilities — let's compare notes.

A stronger test changes both the material and the loading history.

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Does conditioning leave a structural memory?

└ Copper establishes the effect, not universality

The honest limitations: one conditioned cathode and one external reference; field level and breakdown density covary; LAM is a relative proxy. Invite collaboration on replication and cross-facility comparisons. Depth matters because the STEM depletion was seen in hard copper and the EBSD curvature in heat-treated copper, each in the top few hundred nanometres at most; a depth profile on one material would show whether they are two ends of one redistribution. The next experiments therefore add depth, and change the material and loading mode.

Copper establishes the effect, not universality	
What is established	What remains open
▶ Large-area structural difference ▶ Three independent metrics ▶ Internal and external controls ▶ Confirms the MDDF prediction	▶ One conditioned cathode / one experiment ▶ Full depth profile ▶ Grain-orientation dependence ▶ More conditioned cathodes ▶ Cross-material generality
Call for cooperation: more cathodes, other materials, other facilities — let's compare notes. A stronger test changes both the material and the loading history.	

Next: go deeper, and change the material



Depth

- ▶ Matched high-field / reference pilot FIB lamellae
- ▶ Artifact controls learned from the 2025 “red herrings”
- ▶ Correlated TEM/STEM + EBSD through the top $\sim 1\text{ }\mu\text{m}$
- ▶ Do hard-Cu depletion (STEM) and soft-Cu curvature (EBSD) form one depth profile?

Depth continues the Uppsala STEM line; material tests the mechanism itself.

Material & loading

- ▶ Stainless steel: lower stacking-fault energy
- ▶ Different dislocation mobility and slip
- ▶ Continuous DC instead of pulsed loading
- ▶ Possible metastable phase transformation

Source: Interim work plan (2026); RFX project plan.

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└─ Next: go deeper, and change the material

Two orthogonal directions. Depth is a direct continuation of the STEM work: pilot lamellae from a mapped high-field ROI and a matched reference, with the artifact controls we learned we need in 2025. The material axis is the RFX stainless-steel collaboration, which changes stacking-fault energy, dislocation mobility, and the loading mode from pulsed to continuous DC.

Next: go deeper, and change the material

Depth	Material & loading
▶ Matched high-field / reference pilot FIB lamellae ▶ Artifact controls learned from the 2025 “red herrings” ▶ Correlated TEM/STEM + EBSD through the top $\sim 1\text{ }\mu\text{m}$ ▶ Do hard-Cu depletion (STEM) and soft-Cu curvature (EBSD) form one depth profile?	▶ Stainless steel: lower stacking-fault energy ▶ Different dislocation mobility and slip ▶ Continuous DC instead of pulsed loading ▶ Possible metastable phase transformation

Depth continues the Uppsala STEM line; material tests the mechanism itself.

Source: Interim work plan (2026); RFX project plan.

Stainless steel: a test chosen to be hard on the model



Copper
FCC, higher SFE
pulsed DC, $\sim 10^9$ cycles

AISI 304L SS
metastable austenite, lower SFE
continuous DC, possible $\gamma \rightarrow \alpha'$

SS tracks field supports a cross-material structural state

SS differs identifies material-specific carriers
no resolved change bounds depth, dose, sensitivity

Design: apex-to-periphery gradient; two exposed + two matched references.

Status: 90 h and 73 h of DC at up to 61 MV/m, no breakdowns; EBSD indexes three of four samples, one field-exposed sample still does not.

In collaboration with 
Consorzio RFX — A. De Lorenzi, N. Pilan, E. Spada

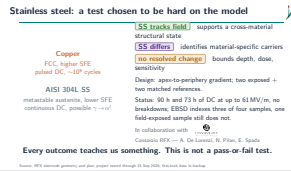
Every outcome teaches us something. This is not a pass-or-fail test.

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Does conditioning leave a structural memory?

└ Stainless steel: a test chosen to be hard on the model

Stainless steel is chosen because it is hard: it changes the stacking-fault energy, the dislocation mobility, the loading mode, and adds a possible austenite-to-martensite transformation. That is what makes it discriminating. A field-correlated response supports a broad structural mechanism; a qualitatively different response identifies which material degrees of freedom store the state; a well-quantified null result bounds the sensitivity and may separate cyclic from continuous loading. Where we stand: the two exposed electrodes ran about 90 and 73 hours of continuous DC at up to 61 megavolts per metre with no full breakdowns; three of the four samples give indexable EBSD, and one field-exposed sample still does not, which is a sample-preparation problem, not a field effect. An early apex-versus-side contrast turned out to reproduce on a never-installed reference, so it is forming geography. No stainless-steel field effect is claimed yet. First-look SEM and EBSD are in backup.

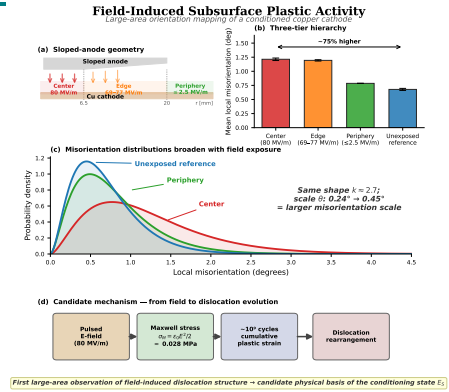


Conditioning appears to leave a subsurface memory

- 1. Our model predicts conditioning evolves the near-surface dislocation structure.
- 2. 2025 STEM saw it locally; **large-area EBSD now confirms it** (~75% higher misorientation).
- 3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.

Last year: “more samples? other materials?” — this is the large-area answer.

Copper result: PRAB, under revision; arXiv:2606.19192.



2026-09-28

Does conditioning leave a structural memory?

Conditioning appears to leave a subsurface memory

Close the loop from the title and from last year. The model predicted a structural memory; the STEM saw it at one spot; the large-area EBSD confirms it across the electrode with a 75 percent effect and a three-tier hierarchy. The stainless-steel and depth programs are the next round of comparative forensics I promised in 2025. Invite collaboration on more cathodes and cross-material tests. Cite the manuscript as under revision at PRAB, arXiv:2606.19192.

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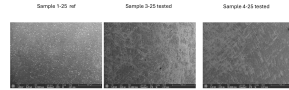
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Backup: RFX first look (preliminary)



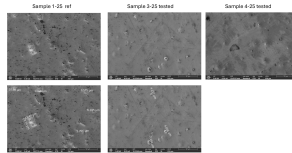
Preliminary — no field effect claimed

- ▶ AISI 304L at RFX HVPTF; continuous DC: 90 h (#3-25) and 73 h (#4-25), peak 59–61 MV/m, no full breakdowns



Sectioned electrode

Ref vs field-exposed overview



- ▶ As-received SEM: “exploded” near-surface inclusions (3–7 μm) on field-exposed samples; reference cavities round and clean

- ▶ Plan-view EBSD: #1-25, #2-25, #3-25 index at 97–99.5%; #4-25 poor patterns, left unpolished

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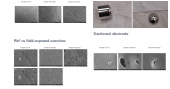
└ Backup: RFX first look (preliminary)

Backup for questions on the stainless-steel first look. SEM survey plus EBSD quality: three of four samples index; the field-exposed #4-25 does not, and we decided not to repolish it. The July apex-versus-side contrast reproduced on a never-installed reference, so it reflects forming geography. No structural field effect is claimed.

Backup: RFX first look (preliminary)

Preliminary — no field effect claimed

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Backup: RFX sample and data status (15 Sep 2026)



Sample	Role	Exposure	Status
#1-25	reference	none	SEM; EBSD apex + side, 97% indexed
#2-25	reference	none	EBSD apex (18 Aug, 3 maps) + side; LAM done
#3-25	field-exposed	90 h DC, peak 61 MV/m	SEM; EBSD apex + side, 99.5% indexed; LAM
#4-25	field-exposed	73 h DC, peak 59 MV/m	SEM/EDS; EBSD patterns poor, not repolished

- Received since July: conditioning report and $E(\theta)$; alloy confirmed 304L (certificate unrecoverable); electropolishing partly documented; five per-pixel maps.
- Still open: raw patterns unavailable; per-pixel data for the wide-grain class; matched EDS overlay checks.

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Does conditioning leave a structural memory?

Backup: RFX sample and data status (15 Sep 2026)

Use only if asked about readiness or missing inputs. Reflects the written project record through 15 September 2026.

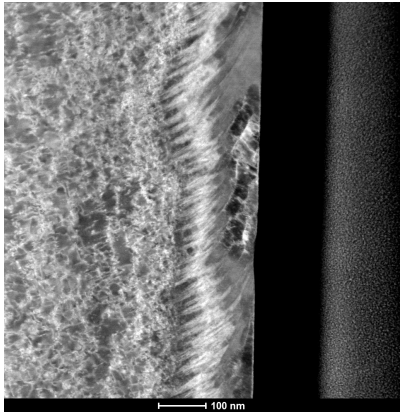
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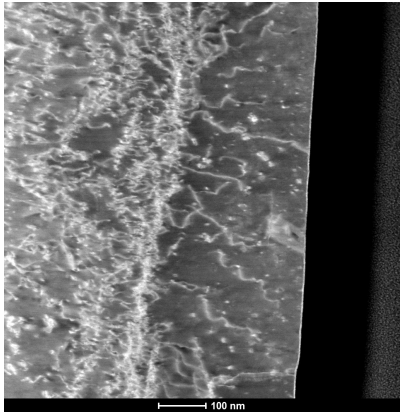
Still open: raw patterns unavailable; per-pixel data for the wide-grain class; matched EDS overlay checks.

Backup: field-exposed vs reference STEM (2025)



Field-exposed — denuded near surface

Source: Jacewicz et al., JAP 137, 193302 (2025).



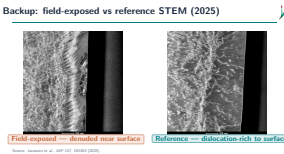
Reference — dislocation-rich to surface

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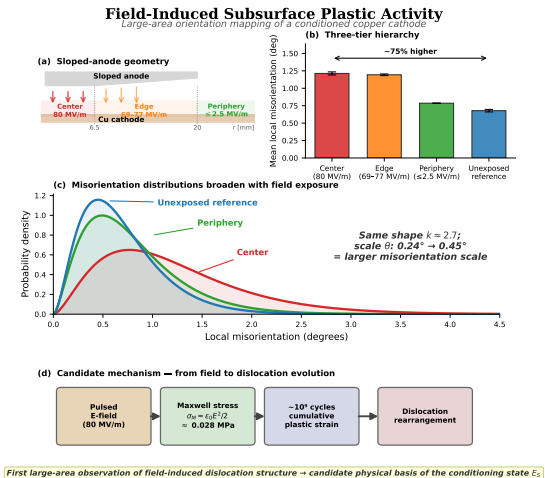
Does conditioning leave a structural memory?

└ Backup: field-exposed vs reference STEM (2025)

The direct field-exposed vs reference STEM comparison behind the denuded-zone claim.



Backup: copper result in one figure



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Does conditioning leave a structural memory?

Backup: copper result in one figure

Compact answer to questions about geometry, distributions, or the proposed mechanism.

